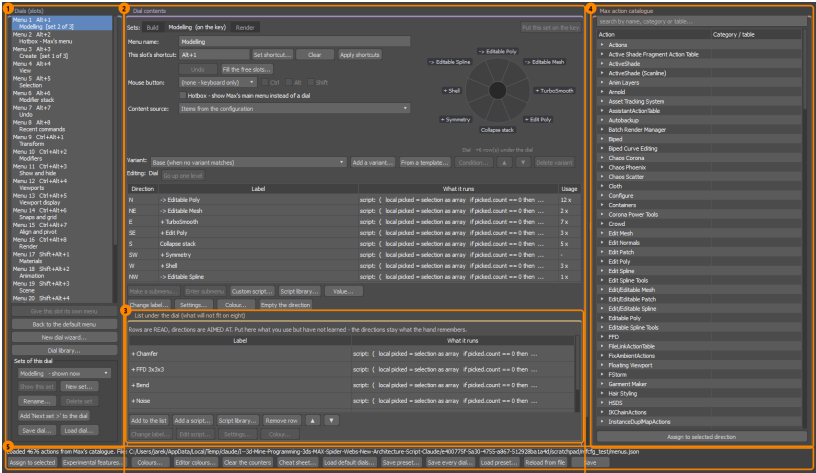


Editor Guide - your first dial, and every part of the editor



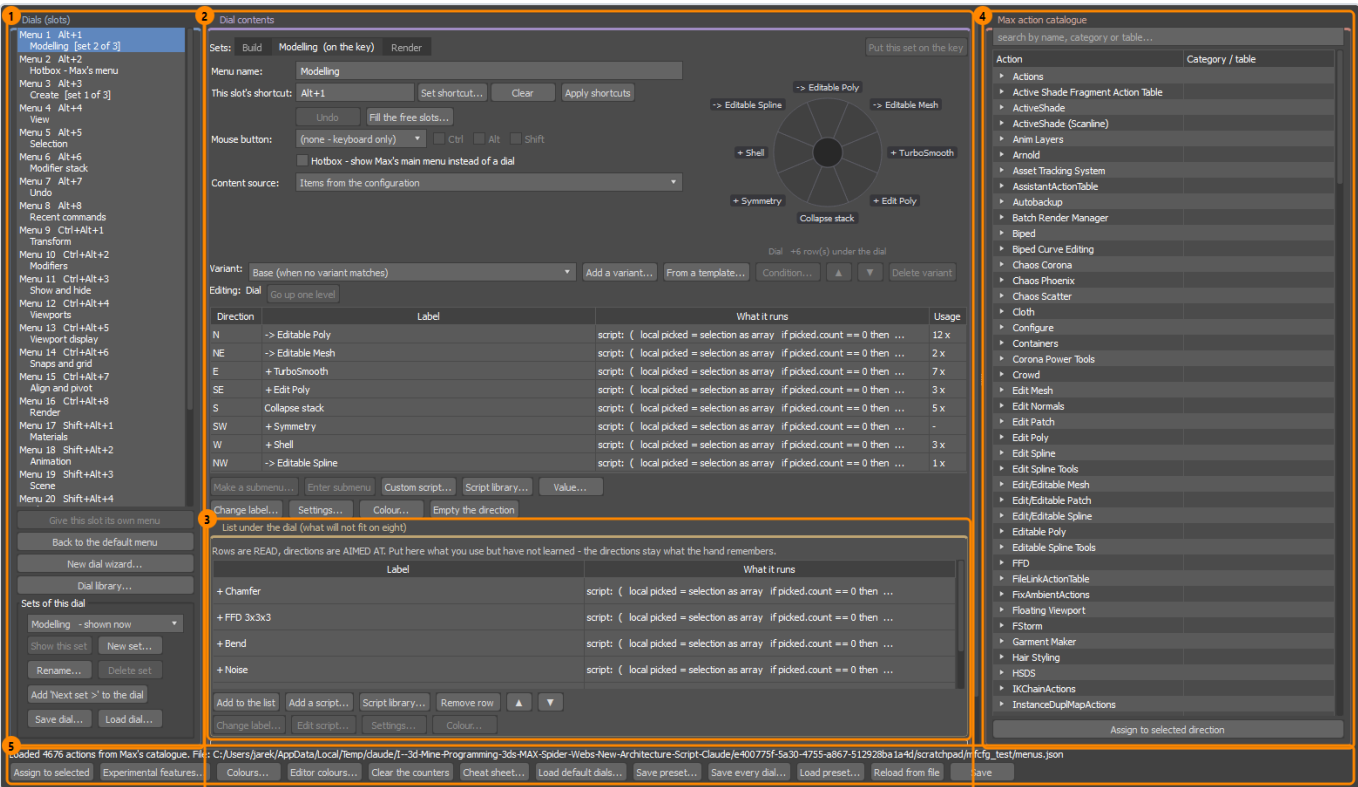
The menu editor decides what every dial holds: the commands on its eight directions, the rows under it, its key, what it shows in which situation and how it looks. This guide builds a dial from start to finish, then walks through the rest of the window piece by piece. The **QuickStart** teaches the gestures; the **Reference** lists every option.

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7. Submenus
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1. The editor at a glance

Open it with **MarkingForge > Menu editor...** Four panes on a neutral grey, each marked by its own colour along its top edge and in its title, and a bar of commands along the bottom:



The editor. The colour bar along each pane's top edge tells the panes apart.

	Pane	What you do there
	Dials (slots) - left	Pick which of the 24 dials you are editing. Each line shows the dial's key and its name. Below the list: the New dial wizard , the Dial library and the dial's sets (chapters 12 and 13).
	Dial contents - middle	Name, key, mouse button, what the dial shows (content source) and, beside them, a drawing of the dial ; its variants, and the table of eight directions with the buttons that change them.
	List under the dial - inside the middle	Up to twelve extra rows drawn under the dial, with their own buttons.
	Max action catalogue - right	Every command 3ds Max and its plugins offer, searchable. Drag from here onto a direction or a row.
	Bottom bar	Save, Reload from file, the dials' colours and the editor's own, experimental features, presets, the cheat sheet.

Nothing reaches the dials until you press Save. Experiment freely; **Reload from file** throws every unsaved change away. Saving reloads the dials in about 150 ms - no need to restart 3ds Max.

2. Your first dial, start to finish

Every one of the 24 slots comes with a built-in dial, so "your own dial" means taking over a slot. This chapter takes **Menu 16 (Render)** - pick any slot you use least - and turns it into *My tools*: a few commands from the catalogue, a ready-made script, a row under the dial and a key of your choice.

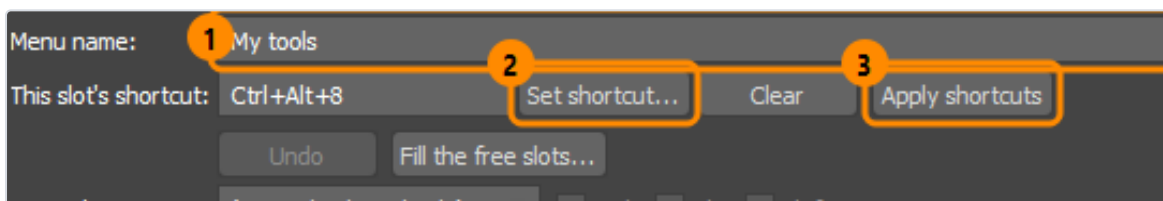
In a hurry? New dial wizard... under the slot list asks three questions - which dial, which eight commands, done - and keeps what the dial showed as another set (chapter 13). This chapter shows the whole road, which the wizard shortens.

1

Pick the slot and name it

1 min · You will learn: where a dial's name shows

- 1 In **Dials (slots)** click **Menu 16**.
- 2 In **Menu name** type *My tools*. The name appears in the dial's footer, in the cheat sheet and in 3ds Max's Hotkey Editor next to the slot.



Menu name (1), the slot's shortcut and Set shortcut... (2), Apply shortcuts (3).

2

Give it a key

1 min · You will learn: to open the dial with any keys you like

- 1 Press **Set shortcut...**, then press the keys - for example **Shift+Q**. Backspace clears, Esc cancels.
- 2 If the keys already open another dial, the window says which one; OK moves them here, and that dial is left without a key until you give it one.
- 3 Press **Apply shortcuts**: only then does 3ds Max learn the new key. **Undo** next to it restores the keys from before.

3

Put commands on directions

3 min · You will learn: the one skill everything else builds on

- 1 Select a direction in the table - say **N**.
- 2 In the catalogue on the right type part of a command's name - *extrude*.
- 3 Drag the command onto the direction, or double-click it, or press **Assign to selected direction**.
- 4 Do the same for a few more directions. Think in pairs: opposite directions for opposite commands (hide north, unhide south) are the easiest to remember.

Direction	Label	What it runs	Usage
N	-> Editable Poly	script: (local picked = selection as array if picked.count == 0 then ...	12 x
NE	-> Editable Mesh	script: (local picked = selection as array if picked.count == 0 then ...	2 x
E	+ TurboSmooth	script: (local picked = selection as array if picked.count == 0 then ...	7 x
SE	+ Edit Poly	script: (local picked = selection as array if picked.count == 0 then ...	3 x
S	Collapse stack	script: (local picked = selection as array if picked.count == 0 then ...	5 x
SW	+ Symmetry	script: (local picked = selection as array if picked.count == 0 then ...	-
W	+ Shell	script: (local picked = selection as array if picked.count == 0 then ...	3 x
NW	-> Editable Spline	script: (local picked = selection as array if picked.count == 0 then ...	1 x

2 Make a submenu... Enter submenu Edit script... Script library... Value...
 3 Change label... Settings... Colour... Empty the direction
 List under the dial (what will not fit on eight)

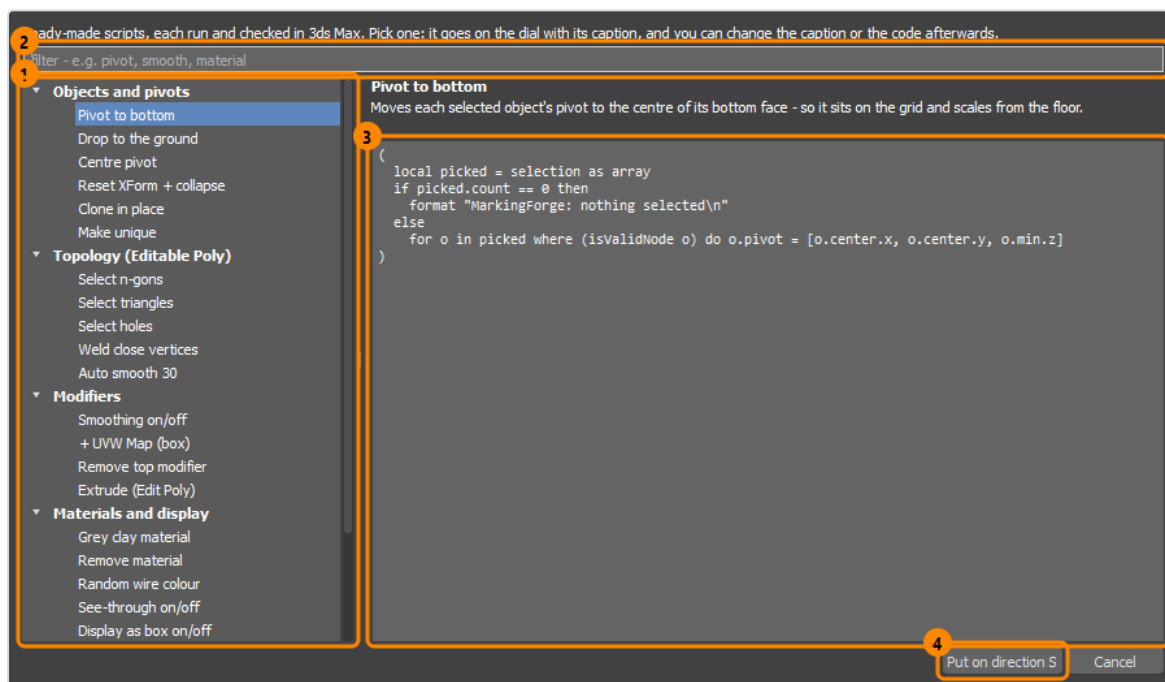
The directions (1) with their buttons: what a direction IS (2), how it LOOKS (3).

4

A ready-made script

1 min · You will learn: commands the catalogue does not have

- 1 Select an empty direction and press **Script library...**
- 2 Pick **Pivot to bottom** (or any other) and press **Put on direction**.



The script library.

5

Short captions

1 min · You will learn: to keep the dial readable

- 1 Select a direction with a long name and press **Change label...**
- 2 Type two words at most - *Extrude* rather than *Extrude Polygons (Editable Poly)*.

6

More than eight: a row under the dial

1 min · You will learn: to keep rarely used commands on the same key

- 1 Find a command in the catalogue and press **Add to the list** in the list's bar.
- 2 Rows are read and clicked, directions are aimed at: put here what you use but have not learned.

7

Save and try it

1 min · You will learn: to see the dial

- 1 Press **Save**.
- 2 Back in 3ds Max hold your key: *My tools* opens with your commands. Flick towards one to run it without waiting for the dial.

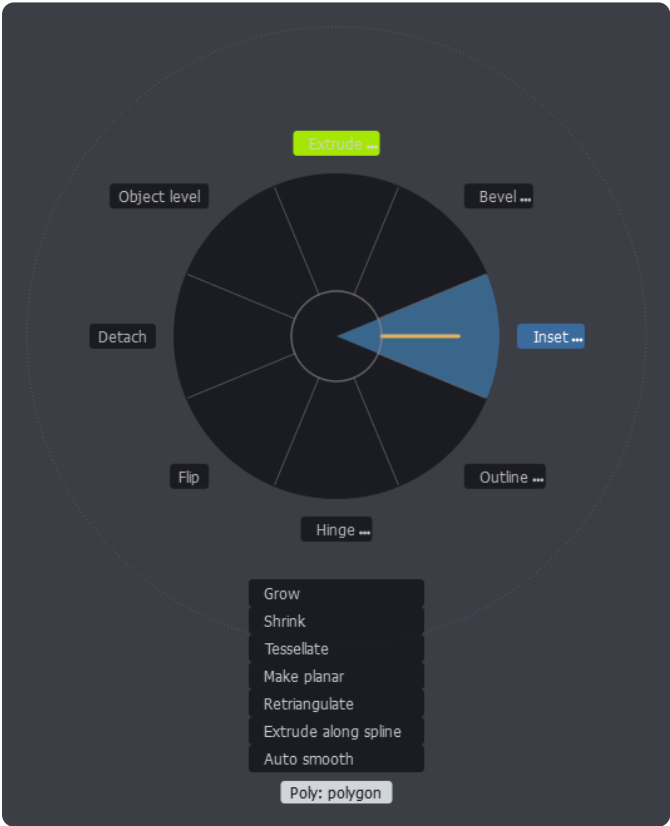
RESULT

A dial of your own, on your key. Everything below makes it smarter: different contents for different objects, the caddy past the rim, submenus, live contents, colours.

3. Rings and circles on the dial

Three round shapes on a dial carry meaning. Each is drawn only when it means something:

You see	It means	Set in the editor
The circle in the centre (36 px across its radius)	Release here to cancel - always, whatever else is going on. It turns to the cancel colour when a release would do nothing. Since 1.0.9 it is twice the size it was, so the hand finds it without aiming.	-
A dotted circle outside the dial (the settings rim)	Commands with a caddy (Chamfer, Extrude, Inset...): an ordinary release does one thing, releasing past this rim the opposite - with or without the caddy. It brightens once you are past it. It lies 30 px beyond the farthest caption, so a short movement past the captions is enough.	Settings... under the directions; the global tick box in Experimental features
A dashed circle round the centre	Multiple picks: go past the rim and come back inside it and the command waits in a queue (a numbered row above the dial). Releasing in this ring (from the centre circle out to 78 px) - green - runs the queue as one undo; the centre circle still cancels.	Experimental features > Multiple picks in one gesture (off by default)



Inside the rim - an ordinary release.



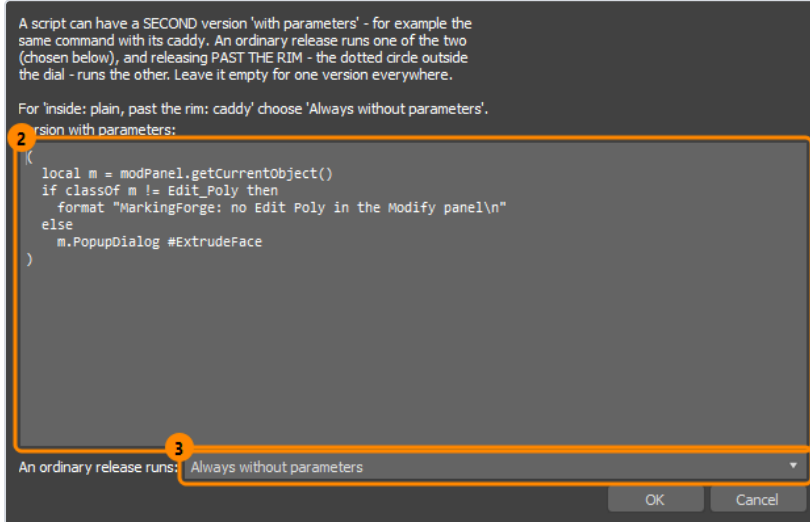
Past the rim - the opposite choice.

The settings rim, step by step

- 1 Select a direction with a command that has a caddy - *Chamfer*, say - and press **Settings...**

- 2 Choose what an ordinary release does: *as the global setting, always with parameters* (the caddy opens) or *always without parameters* (it runs at once).
- 3 Past the rim always does the opposite. Save, then try both in a viewport.

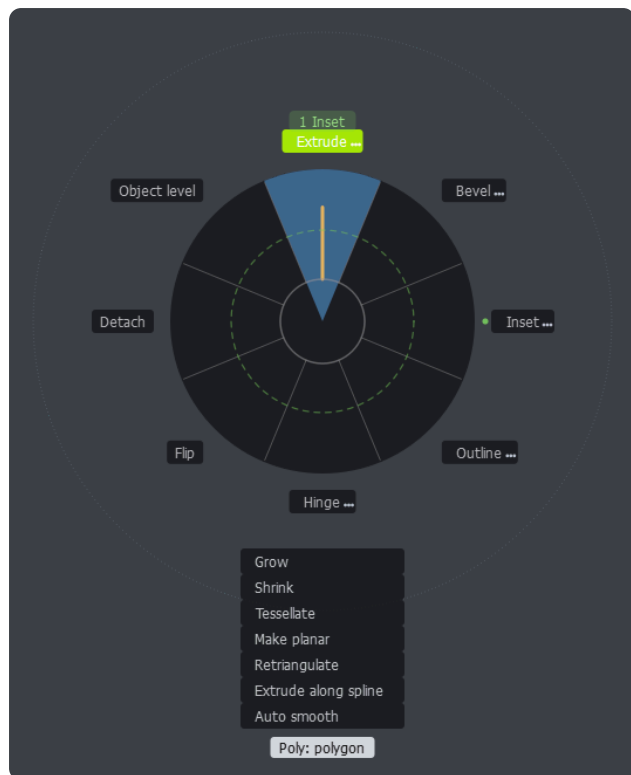
The global setting is the tick box **An ordinary release shows the action's SETTINGS** in **Experimental features** - on by default. For a script, **Settings...** takes a second version of the code instead (the library's *Extrude (Edit Poly)* is one):



Settings... on a script: its version with parameters.

Multiple picks, step by step

- 1 Open **Experimental features...** and tick **Multiple picks in one gesture**; OK, Save.
- 2 Hold a key, move past the rim towards a command, come back to the dashed circle without releasing: it is queued. Do it again for another command.
- 3 Release inside the dashed circle (green): both run, as one undo. Changed your mind? Release on the dot in the middle, or over an empty direction.



A queued command (the row above the dial) and the dashed return circle.

4. Directions, button by button

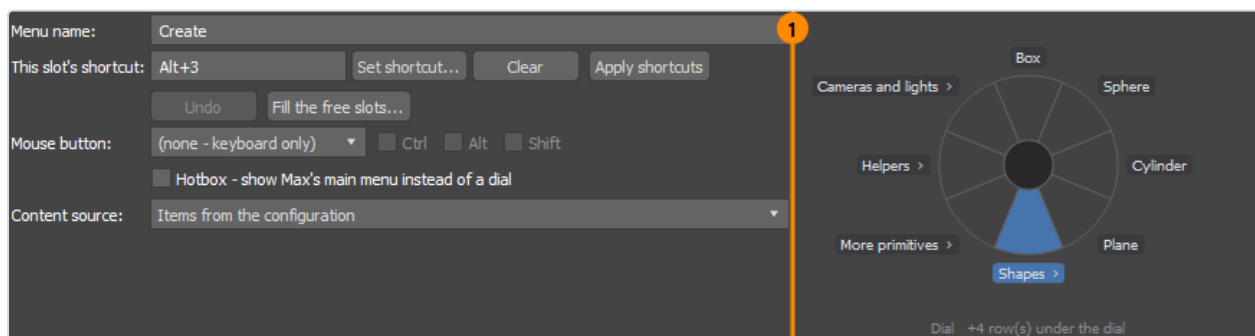
Two rows of buttons under the table. The first changes **what** the selected direction is, the second how it **looks and behaves**.

Button	What it does
Make a submenu...	Turns the direction into another dial with its own eight directions (chapter 7).
Enter submenu	Edits the submenu under the direction. Double-clicking the row does the same.
Custom script... / Edit script...	Your own MAXScript on the direction - or its code, when it already runs a script.
Script library...	57 ready-made scripts, each tested in 3ds Max (chapter 5).
Value...	A slider: the distance you move sets a number (chapter 8).
Change label...	A caption of your own. Empty = 3ds Max's name.
Settings...	With or without the caddy on an ordinary release; for a script, its second version (chapter 3).
Colour...	This one direction's caption and tile colour - a red caption on a destructive command, say.
Empty the direction	Removes what the direction runs; releasing there cancels.

The table shows each direction's caption, what it runs and its **Usage** (chapter 10). A red caption means the command does not exist in this 3ds Max - a missing plugin; replace it.

The drawing of the dial

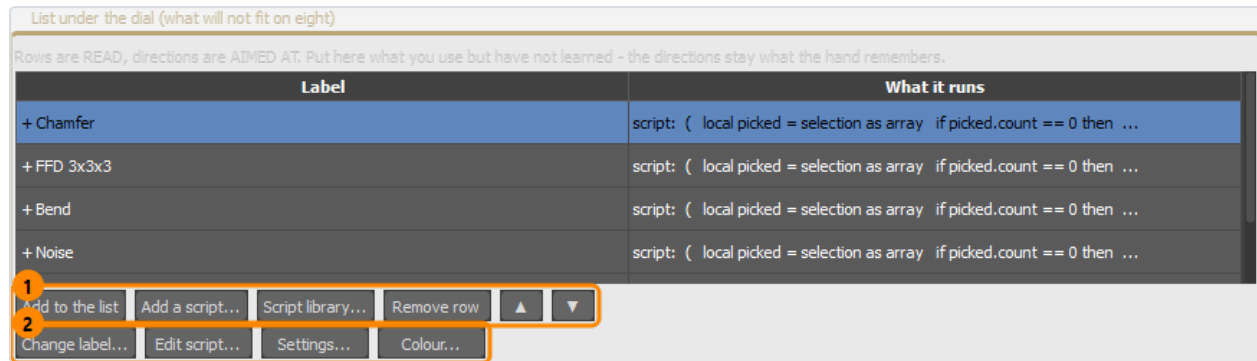
Beside the name and the key the editor draws the dial you are editing - north at the top, as it opens. The table says *what* each direction runs; the drawing says *where*, which is how a marking menu is learnt. **Click** a wedge to select that direction in the table; **double-click** a submenu (marked >) to step inside. A caption in red is a command this 3ds Max does not have. Under the drawing: where you are (dial, variant, submenu) and how many rows the list under the dial holds. In a narrow window the drawing gives way to the form - widen the window or the middle pane to bring it back.



The drawing (1) beside the form: the selected direction in blue, a submenu with >.

5. The list under the dial, and scripts

The list has its own bar, in two rows like the directions' - what is **in** the list above, editing the **selected row** below:



Filling the list (1) and editing the selected row (2).

Button	What it does
Add to the list	Appends the command selected in the catalogue. Dragging onto the list works too - between rows to insert.
Add a script... / Script library...	Appends your own script, or a ready-made one.
Remove row, ▲ ▼	Removes the selected row, moves it up or down. Order is what you see.
Change label..., Edit script..., Settings..., Colour...	The same edits a direction has. A row has no tile, so Colour... sets the caption only.

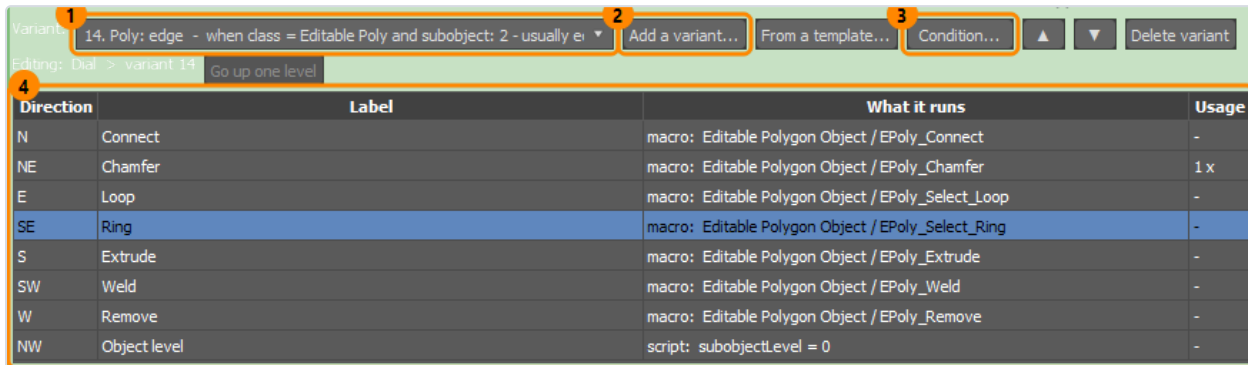
Ready-made scripts

Group	Scripts
Objects and pivots	Pivot to bottom, Drop to the ground, Centre pivot, Reset XForm + collapse, Clone in place, Make unique, Pivot aligned to world, Pivot to world origin, Centre on the origin, Attach to the first, Split into elements
Topology (Editable Poly)	Select n-gons, Select triangles, Select holes, Weld close vertices, Auto smooth 30, Select faces facing up, Select faces facing down, Cap holes, Flip normals, Clear smoothing
Modifiers	Smoothing on/off, + UVW Map (box), Remove top modifier, Extrude (Edit Poly), + Shell, + Symmetry, + Chamfer, + Edit Poly, + FFD 3x3x3
Materials and display	Grey clay material, Remove material, Random wire colour, See-through on/off, Display as box on/off, UV checker material, Material = wire colour, Backface cull on/off
Selection and transforms	Select same material, Select instances, Copy transform, Paste transform, Reset rotation, Select same type, Select children, Select without material, Random rotation (Z), Random scale (10 %), Spread evenly along X
Shapes and splines	Renderable on/off, Close all splines, + Extrude 10
Scene and layers	Selection to a new layer, Freeze selection, Unfreeze all, Zoom to selection, Size to the status bar

Pick one with **Script library...** - the window shows what it does and its code before anything goes on a dial. The Reference prints the code of the most useful four and the shape a script of your own should have; the QuickStart (tutorial 14) puts four of them under the Modelling dial step by step.

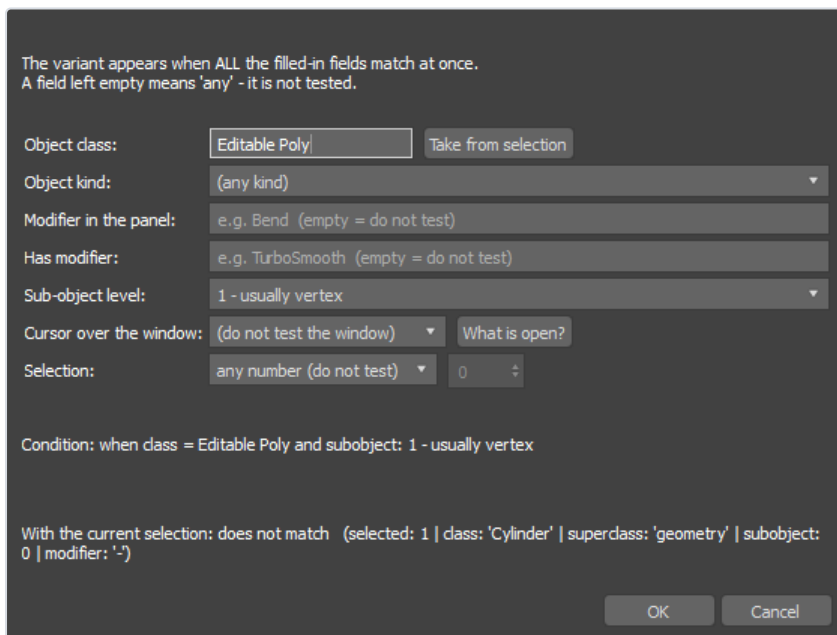
6. Variants - one key, different situations

A dial can have **variants**: the same key shows different contents depending on the object under the cursor (or the selection), its sub-object level, the modifier in the Modify panel, the window under the cursor and more. The built-in Modelling dial has a variant for every Editable Poly level.



Variant (1), Add a variant... (2), Condition... (3), the variant's own directions (4).

- 1 Press **Add a variant...**: it starts as a copy of what you are looking at, and the condition window opens at once.
- 2 Say when it applies - for example *class: Editable Poly* and *sub-object: Edge* - and press OK. **Condition...** changes it later.
- 3 With the variant selected in **Variant**, fill its directions like any dial; **Menu name** renames it.
- 4 Order matters: the first variant whose condition matches wins. Keep specific ones above general ones (▲ ▼). **Base** is what shows when none matches.



The condition window.

From a template... next to **Add a variant...** adds a ready-made variant instead - one of the built-in dials' context menus with its condition (chapter 13).

7. Submenus

A submenu is a whole dial behind one direction: eight more places, reached in two strokes. Use it for a **group** - all the shapes, all the deformers - when the group is too big for the list and too useful to leave out. On the dial a submenu's caption ends with >.

- 1 Select a direction and press **Make a submenu...**; name it. The editor steps inside.
- 2 Fill its eight directions. **Go up one level** returns. Double-click a submenu - in the table or in the drawing - to step back in.
- 3 On the dial, move THROUGH the submenu direction without releasing: the submenu opens where you are. Up to three levels.



Inside a submenu: the path (1), its directions (2), Go up one level (3).

Submenus in the built-in dials

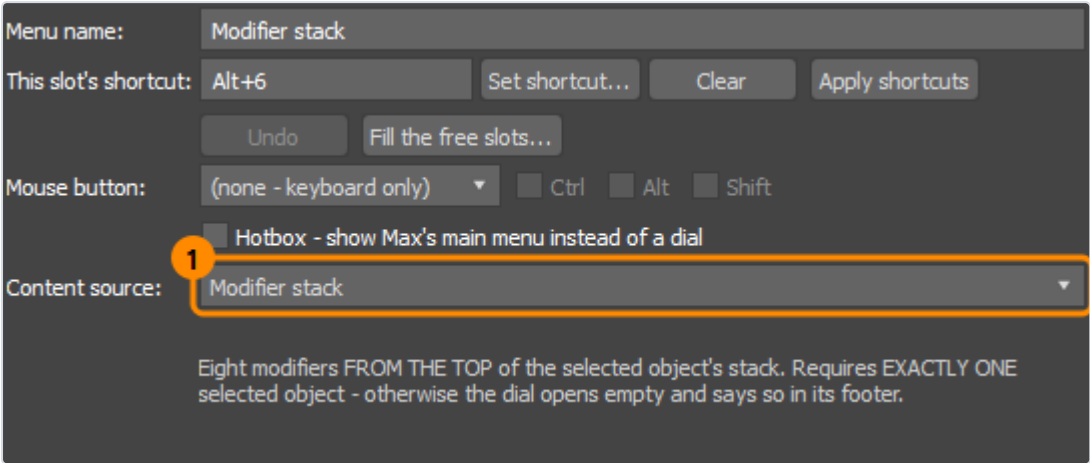
Dial	Direction	Submenu
Alt+3 Create	S, SW, W, NW	Shapes, More primitives, Helpers, Cameras and lights
Ctrl+Alt+2 Modifiers	SW, NW	Deform (Bend, Twist, Taper, Noise, FFD, Relax, Stretch, Skew), Geometry (Mirror, Array, Cap Holes, Vertex Weld, Face Extrude, Quadify Mesh, OpenSubdiv, MeshSmooth)
Alt+1 Modelling, nothing selected	SW, W	Shapes, More primitives - the same submenus as on Create

The Shapes submenu keeps the directions of the Create dial's shape variant (what Alt+3 shows with a shape selected): Line north, Circle east, Rectangle south - a shape is in the same place either way.

Dials you already have keep their layout when MarkingForge is updated - nothing is replaced behind your back. **Load default dials...** brings the current built-in dials in: by default into slots without a menu of their own, or, with its tick box, in place of yours.

8. Content source, sliders, mouse and tap

Content source - dials that fill themselves



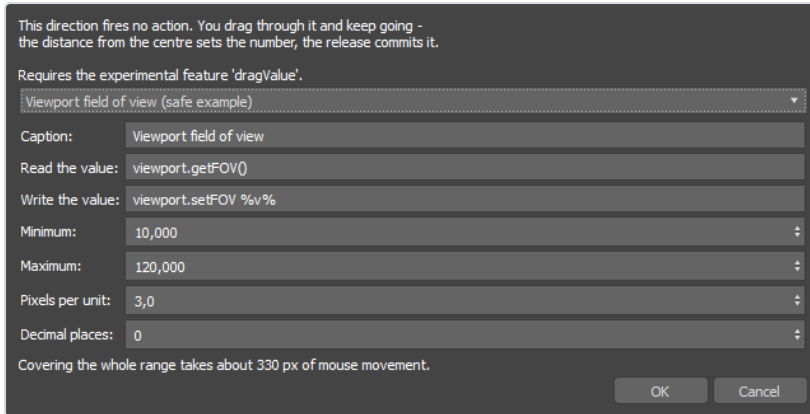
Content source.

Choice	The dial shows	Releasing on a direction
Items from the configuration	Eight directions exactly as you set them here. An ordinary dial.	runs it
Modifier stack	Eight modifiers FROM THE TOP of the selected object's stack. Requires EXACTLY ONE selected object - otherwise the dial opens empty and says so in its footer.	makes it current in the Modify panel
Undo stack	The last eight operations BY NAME. Direction N undoes one, NE two, and so on. Requires the experimental feature 'undoDial'.	undoes back to it
Recent commands	The eight most recently RUN commands, newest first - from any dial, a tap or the search. For REPEATING, as opposed to the undo stack. The history lives in memory and starts afresh when Max starts. Requires the 'recentDial' toggle.	runs it again
Selection sets	The scene's named selection sets as eight directions. With more than eight it shows the first eight. Requires the toggle 'selectionSetsDial'.	selects the set

The QuickStart's tutorial 18 sets up each of them step by step. A source can belong to one variant only - the modifier stack only over the command panel, say.

Value sliders

- 1 Switch on **Experimental features > Direction that sets a value**.
- 2 Select a direction, press **Value...**, pick an example or fill in how to read and write the number.
- 3 On the dial move into that direction and keep going: the distance sets the number, the release commits it.



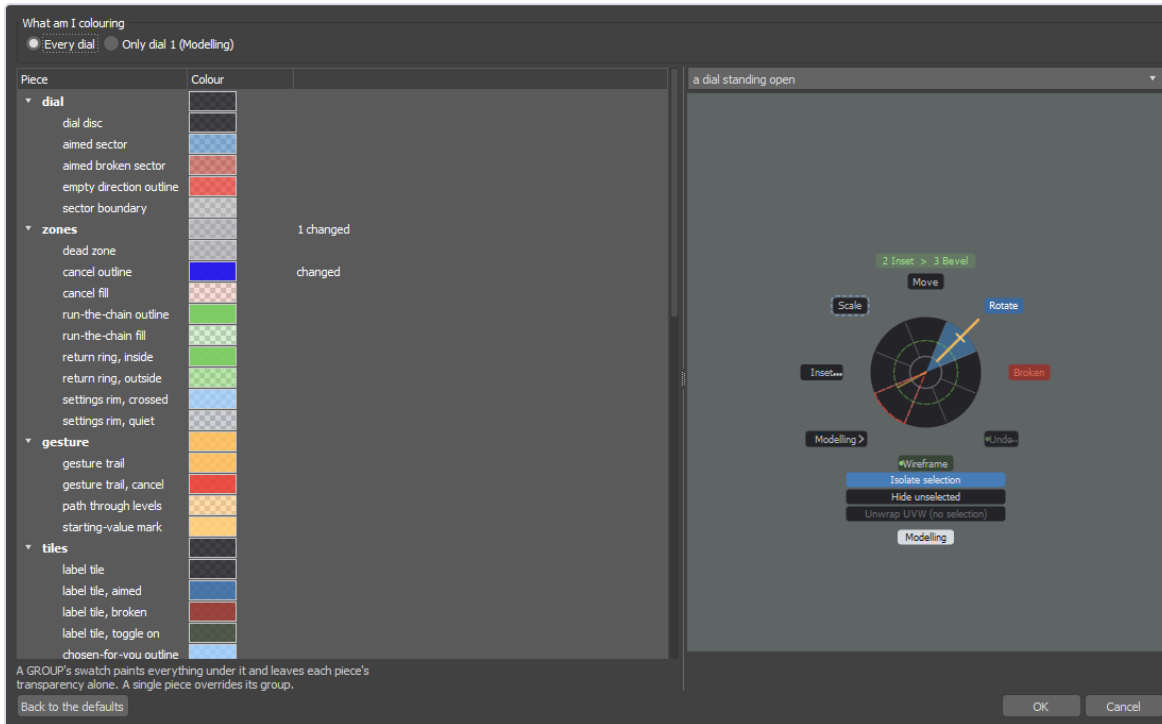
The value window with the viewport field-of-view example.

Mouse button, tap and hotbox

Control	What it does
Mouse button	Opens the dial with a mouse button plus Ctrl, Alt and/or Shift - instead of, or as well as, the key.
A tap runs	What a press-and-release WITHOUT movement does: nothing, or one of the directions.
Hotbox	The tick box shows 3ds Max's whole menu bar as tiles instead of a dial (Alt+2).
Fill the free slots...	Proposes the standard keys (Alt+1-8, Ctrl+Alt+1-8, Shift+Alt+1-8) for every dial that has none, and shows the list - with keys already taken - before writing.

9. Colours

- 1 **Colours...** in the bottom bar sets the palette for all dials or for one dial.
- 2 Click a group (dial, tiles, zones...) or a single piece; the preview shows the dial at rest, aimed, cancelling and chaining.
- 3 OK, then Save.



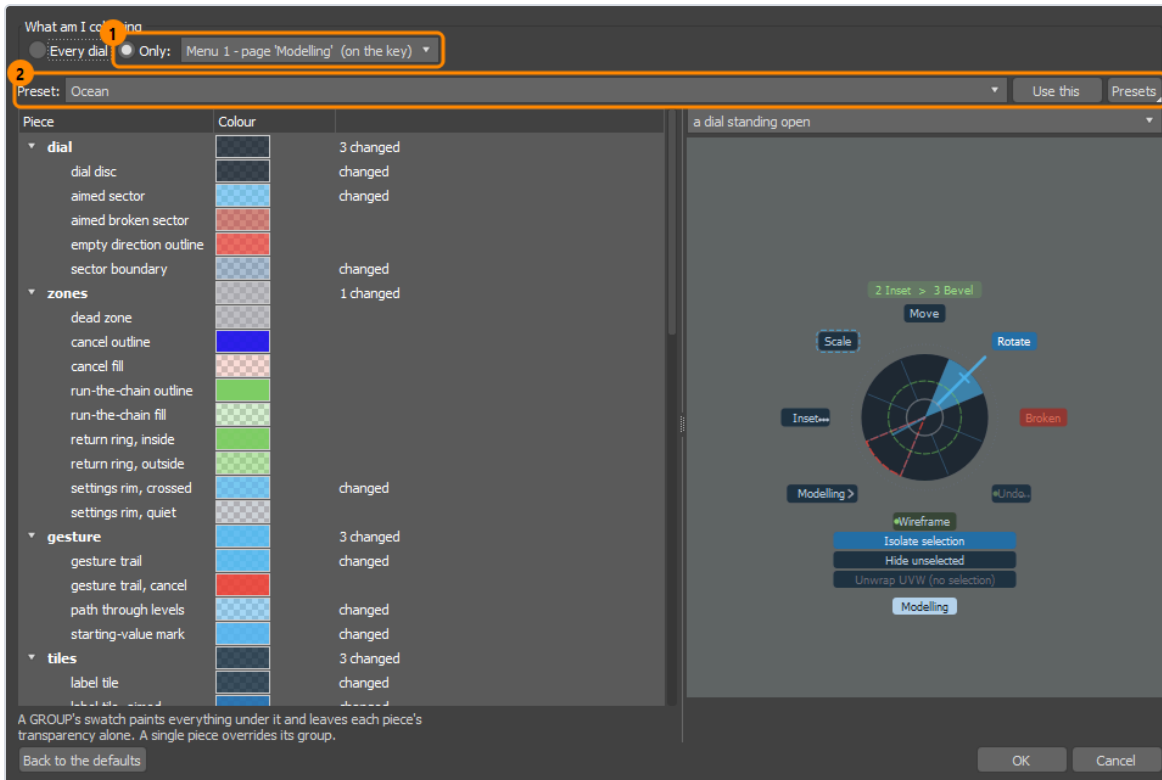
The colour editor.

A colour for one command only: select the direction (or row) and press **Colour...**

Ready-made looks, your own presets - for every dial, one dial or one page

Preset at the top of the colour editor offers 30 ready-made looks - MarkingForge default, Graphite, Slate, Ocean, Violet, Rose, Ember, Mono, Sand - light tiles, High contrast, Indigo, Twilight, Steel, Crimson, Copper, Gold, Ice - light tiles, Lavender - light tiles, Sapphire, Amethyst, Magenta, Coral, Sunset, Amber, Ruby, Storm, Carbon, Neon, Snow - light tiles, Peach - light tiles - and the ones you saved. **Use this** paints what is chosen under **What am I colouring: Every dial**, or **Only** one dial. For a dial with sets the list beside **Only** offers **every page** of it and **each page** on its own - the sets the mouse wheel turns. A dial's colours reach its context variants and its submenus as well.

- 1 Choose the scope: every dial, one dial, every page of it, or one page.
- 2 Pick a preset and press **Use this** - or click swatches.
- 3 **Presets > Save these colours as a preset...** keeps the look under a name; **Export** and **Import** move one to another computer as a `.mfcolors` file.
- 4 OK, then Save.



One page of Menu 1 (1) in the Ocean preset (2).

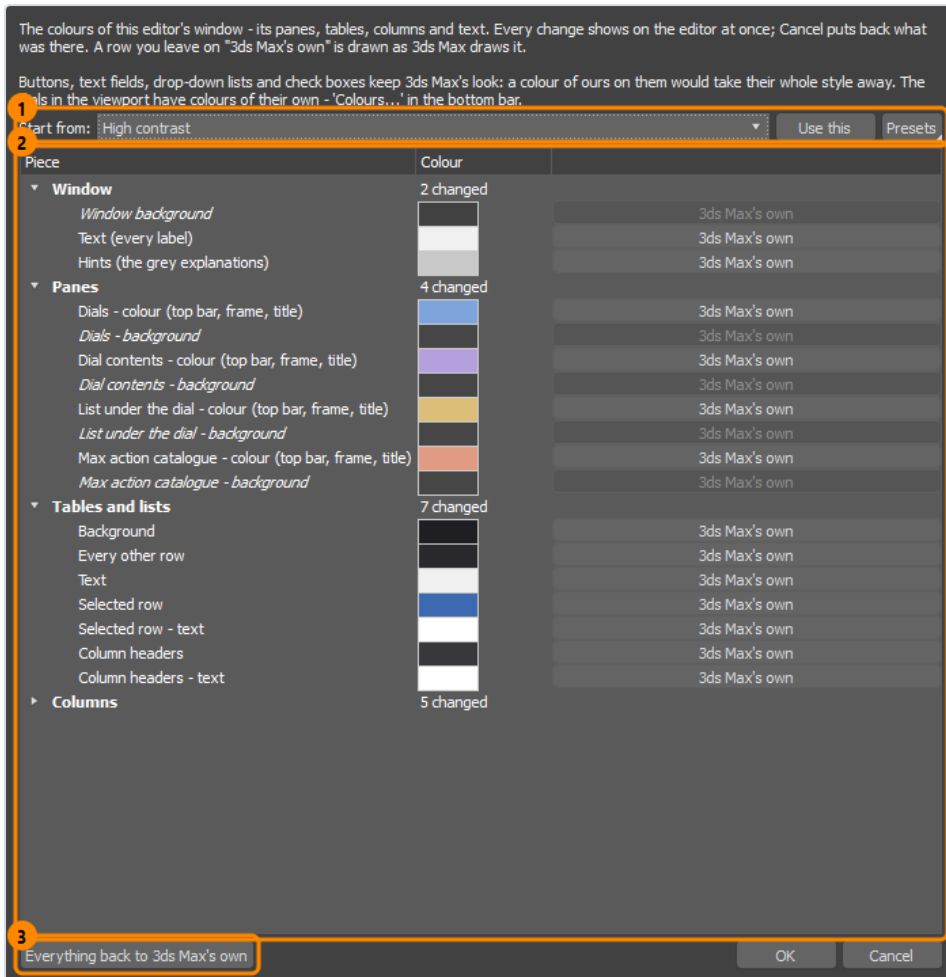


Every ready-made dial look, as the colour editor's preview draws it.

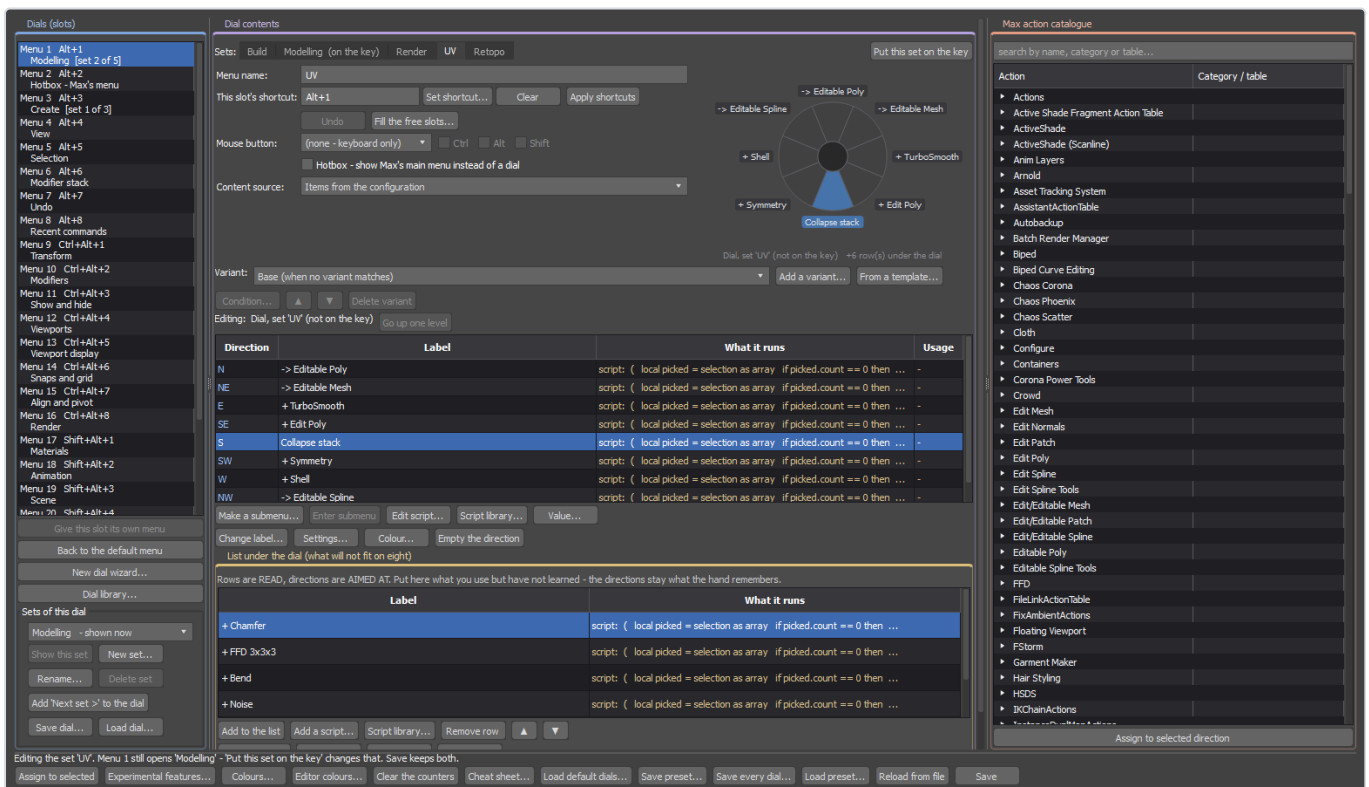
The editor's own colours

Editor colours... in the bottom bar colours this window, not the dials: the window's ground, its text and hints, each pane's colour and ground, the tables and lists - their ground, every other row, text, the selected row and the column headers - and **every column of every table on its own**, text and ground. Nothing is set until you choose it; the editor changes as you pick.

- 1 Pick a starting point under **Start from** - High contrast, Calm grey, Graphite, Midnight blue, Warm sepia, Plum, Ocean, Charcoal, Steel blue, Indigo night, Mocha, Rose dust, Nord, Sapphire, Amethyst, Carbon, Storm, Burgundy, Slate grey, Dusk, Espresso or one of your own - and press **Use this**, or start from 3ds Max's own. **Presets** saves, deletes, exports and imports your own, as for the dials.
- 2 Click any swatch in **Window**, **Panes**, **Tables and lists** or **Columns**.
- 3 **3ds Max's own** beside a row drops that colour; **Everything back to 3ds Max's own** drops them all. **OK** keeps them - no Save needed; **Cancel** puts back what was there.



Editor colours: starting points (1), every piece of the window (2), back to 3ds Max's own (3).



The editor in the High contrast starting point.

Buttons, text fields, drop-down lists and check boxes keep 3ds Max's look - a colour of ours on them would take their whole style away. The colours live in `editor_colors.json` beside `menus.json`; delete that file to start over.

10. Reading the Usage column

The last column of the directions table counts how you use the dial - quick gestures and held dials alike. 12 x means the direction ran twelve times; *cancelled* 3 that you aimed at it three times and nothing ran - on an empty direction, a hint that something is missing there. The counters are kept per variant and per submenu: choose the variant you worked in to see its numbers.

The screenshot displays the MarkingForge 1.4.4 interface. At the top, a table lists directions and their usage. The 'Usage' column is highlighted with an orange box and labeled '1'. Below the table, there are several buttons for editing and managing the directions. A second table, labeled 'List under the dial (what will not fit on eight)', shows a list of directions and their usage. This table is also highlighted with an orange box and labeled '2'. Below the second table, there are buttons for adding and removing rows. At the bottom, a section labeled 'Conclusions' provides a summary of the data.

Direction	Label	What it runs	Usage
N	-> Editable Poly	script: (local picked = selection as array if picked.count == 0 then ...	12 x
NE	-> Editable Mesh	script: (local picked = selection as array if picked.count == 0 then ...	2 x
E	+ TurboSmooth	script: (local picked = selection as array if picked.count == 0 then ...	7 x
SE	+ Edit Poly	script: (local picked = selection as array if picked.count == 0 then ...	3 x
S	Collapse stack	script: (local picked = selection as array if picked.count == 0 then ...	5 x
SW	+ Symmetry	script: (local picked = selection as array if picked.count == 0 then ...	-
W	+ Shell	script: (local picked = selection as array if picked.count == 0 then ...	3 x
NW	-> Editable Spline	script: (local picked = selection as array if picked.count == 0 then ...	1 x

Buttons: Make a submenu..., Enter submenu, Edit script..., Script library..., Value..., Change label..., Settings..., Colour..., Empty the direction

List under the dial (what will not fit on eight)

Label	What it runs
+ Chamfer	script: (local picked = selection as array if picked.count == 0 then ...
+ FFD 3x3x3	script: (local picked = selection as array if picked.count == 0 then ...
+ Bend	script: (local picked = selection as array if picked.count == 0 then ...
+ Noise	script: (local picked = selection as array if picked.count == 0 then ...

Buttons: Add to the list, Add a script..., Script library..., Remove row, ▲, ▼, Change label..., Edit script..., Settings..., Colour...

Conclusions:

- Never used across 33 picks: SW. Space on a dial is limited to eight items.
- More dead-zone cancels (66) than picks (33) - check whether the shortcut is firing by accident.

The Usage column (1) and the conclusions drawn from it (2).

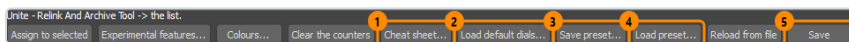
Under the table the editor writes **conclusions** once it has enough data - a direction never used, cancels outnumbering picks. It never rearranges anything by itself: the hand remembers directions, and a dial that reorders itself would undo that. The editor reads the counters when it opens; **Reload from file** refreshes them, **Clear the counters** starts afresh.

11. Experimental features and the bottom bar

Experimental features... switches features on and off. Each states in the window why it is experimental.

Out of the box:

Feature	Default
Direction that sets a value (drag further = more)	off
Typing with the menu open searches for actions	off
Menus loaded from the scene file	off
Multiple picks in one gesture	off
Undo dial	on
Menu under the cursor	on
Recent commands dial	on
Selection sets dial	on
Modifier under the cursor	on
Operation-pair learning	on
Chain as an item	on

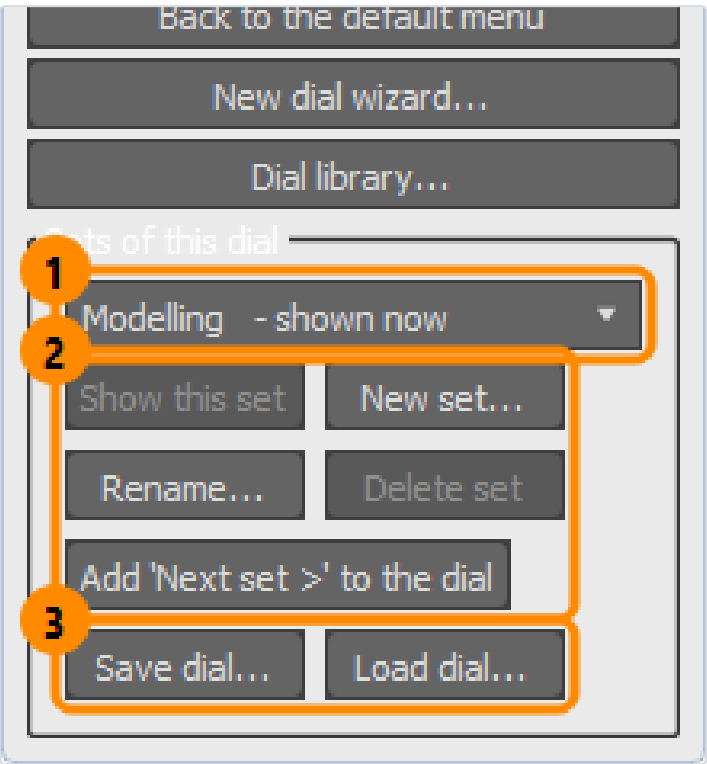


Cheat sheet (1), Load default dials (2), Save preset (3), Load preset (4), Save (5).

Button	What it does
Save	Writes the dials; the plugin reloads them at once.
Reload from file	Throws unsaved changes away and re-reads the counters.
Cheat sheet...	A printable PDF of your saved dials with their keys.
Save preset... / Load preset...	Your layout in one file, to share or keep. Loading shows what a file holds first and removes scripts from other people's presets by default.
Save every dial...	Each of the 24 dials - and every other set of a dial - in a file of its own, in a new dated folder. Load dial... puts any one back alone.
Load every dial...	Every dial of such a folder back at once: what each dial showed, its sets in their order, the dials that showed the default menu and the colours of every dial. Scripts from the files are removed unless you keep them; nothing is written until Save. Right-click a dial - in the list of dials or on its drawing - for this, Save every dial... and saving or loading that one dial.
Load default dials...	The built-in layout back - by default only for dials without their own menu.
Clear the counters	Zeroes the Usage column.

12. Sets - several dials on one key

A dial can keep several **sets** of contents under names - *Modelling, UV, Retopo* - and switch between them in place. The set the dial shows is an ordinary dial; the others wait beside it in the same file. The key, the mouse button and the other 23 dials stay as they are.

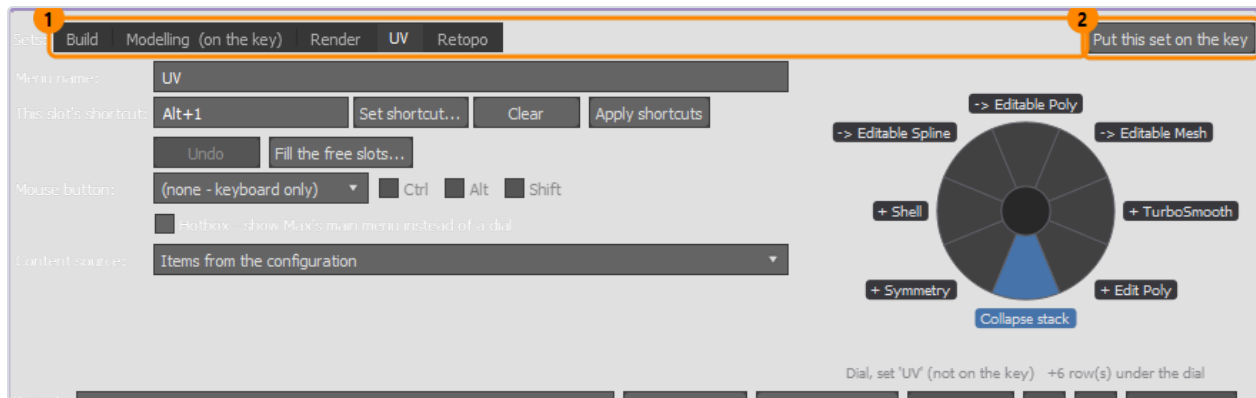


Sets of this dial: the sets (1), switching and managing them (2), one dial in a file (3).

Button	What it does
New set...	Adds a set - a copy of what the dial shows now, or an empty dial - and opens its tab. The key keeps opening what it opened; the new set is the dial's next page.
Show this set	Makes the set chosen in the list the one the dial shows.
Rename... / Delete set	Renames the chosen set; deletes it. The set the dial shows cannot be deleted - show another one first. With one set left, the dial is simply a dial again.
Add 'Next set >' to the dial	Puts a <i>Next set ></i> row on the list under the dial in every set (and their variants), so you can switch from the dial and come back.
Save dial... / Load dial...	One dial in a .mfdial file: save the set chosen in the list; load a file as a new set or in place of what the dial shows. Scripts in a file from somebody else are removed unless you choose to keep them.

Looking at another set - tabs and the mouse wheel

Over the dial's contents each set is a **tab**; the one the key opens is marked *on the key*. Click another tab to see and edit that set - the drawing, the directions, the list and the variants are all that set's - while the key keeps opening the set it opened. **Put this set on the key** changes that.



The sets as tabs (1): another set open for editing, and the button that puts it on the key (2).

In 3ds Max the sets are the dial's **pages**: hold its key and **turn the mouse wheel** - the dial shows the next set (wheel towards you) or the previous one, at the same place. Release over a command to run it. The next press opens the set on the key again, so the hand always finds the same dial first. The footer says *page 2 of 3 - mouse wheel*. The wheel is taken only while such a dial is open; at any other time it zooms and scrolls as always.

Ways to switch the set on the key

Where	How
In the editor	The set's tab and Put this set on the key - or pick it in Sets of this dial and Show this set . Save.
On the dial	The <i>Next set</i> > row (see above). The dial changes in a moment and the status bar says which set it shows - <i>Menu 1 -> set 'UV' (2 of 3)</i> .
Under a key	Customize > Hotkey Editor , category MarkingForge: Menu 1 - next set ... Menu 24 - next set . Give the one you use a key.

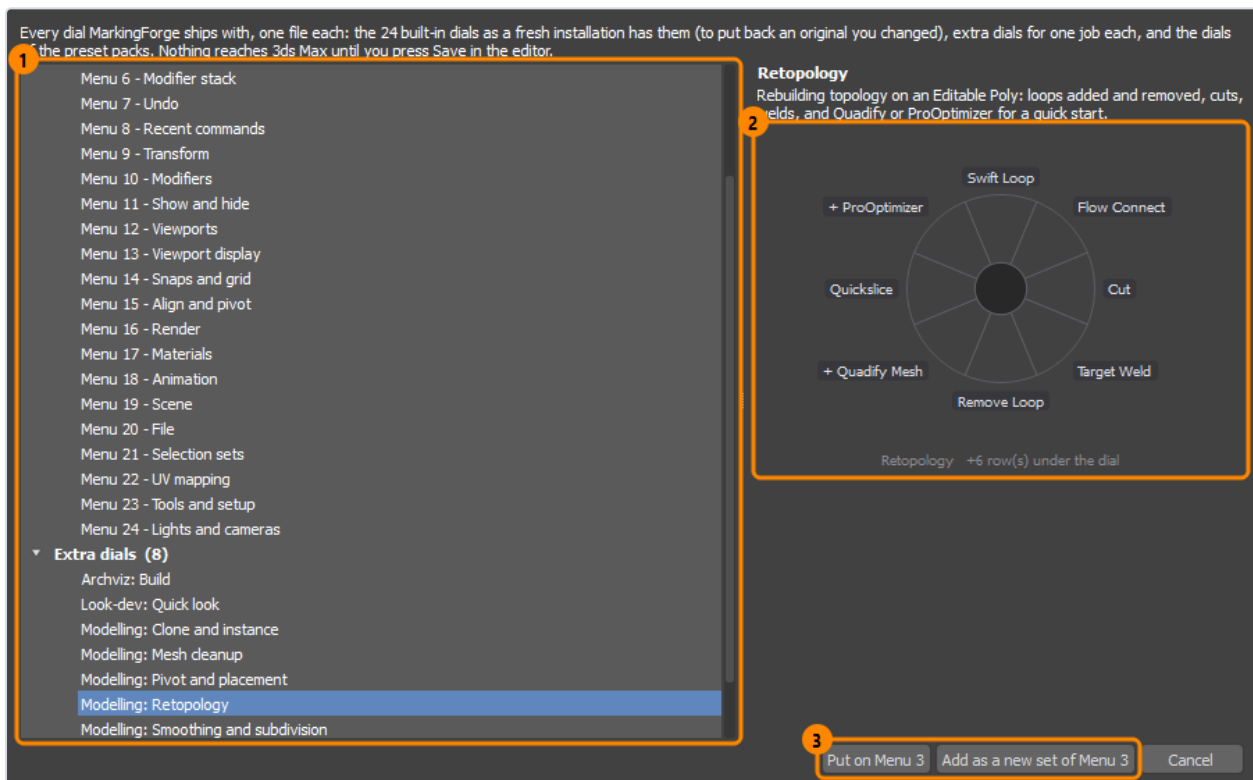
A switch from the dial or a key writes the file at once. If the editor is open with unsaved changes, press **Reload from file** before you go on - Save would find the file changed and offer a separate copy instead of overwriting the switch.

13. The quick ways: the library, the wizard and ready-made variants

The dial library - every dial as a file of its own

MarkingForge ships every dial as a separate `.mfdial` file: the **24 built-in dials** exactly as a fresh installation has them, **extra dials** for one job each, and the dials of the **preset packs** one by one. **Dial library...** under the slot list opens them, with a drawing of each.

- 1 Pick the dial you want to change in the slot list and press **Dial library...** - its built-in original is already selected.
- 2 **Restore Menu N** puts that original back - after you changed it and want it as it was.
- 3 Or pick any other dial on the left and press **Put on Menu N** (in place of what the dial shows) or **Add as a new set of Menu N** (the key keeps opening what it opens; the new set is its next page - the mouse wheel - and its tab opens here, chapter 12).
- 4 Save.



The dial library: built-in dials as shipped, extra dials and the packs' dials (1), the chosen dial (2), restore, put on the dial or add as a set (3).

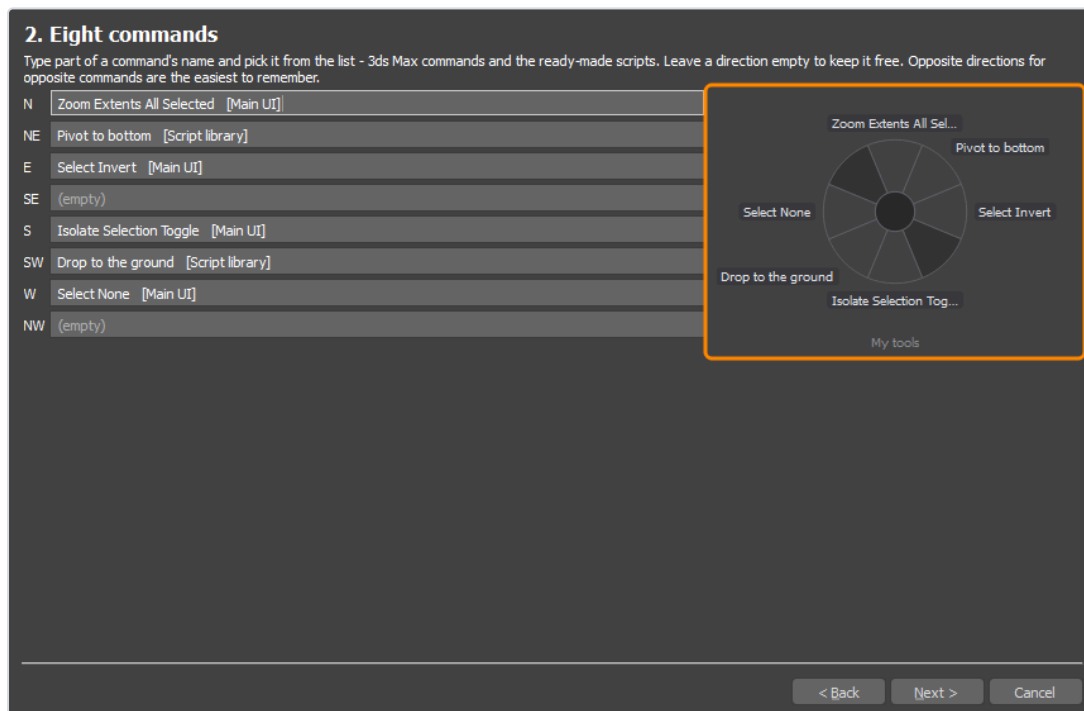
Extra dial	What it is for
Modelling - Mesh cleanup	Find and fix what breaks smoothing and export: n-gons, triangles and holes on one side, the fixes on the other; reset the transform; reduce.
Modelling - Pivot and placement	Where an object's pivot is and where the object stands: pivot to the bottom or the centre, onto the ground, copy a transform from one object to another, align.
Modelling - Clone and instance	Every way to make more of an object: clone, array, spacing along a path, snapshot, mirror - and the instance tools on the other side.
Modelling - Smoothing and subdivision	Subdivision modifiers north and east, smoothing fixes south and west; switch every smoothing modifier of the selection off and on in one move.

Modelling - Retopology	Rebuilding topology on an Editable Poly: loops added and removed, cuts, welds, and Quadify or ProOptimizer for a quick start.
Look-dev - Quick look	Checking a model at a glance: grey clay, no material, random wire colours, see-through, box display and the UV checker.
Rigging - Links and helpers	Building a hierarchy: link and unlink, walk up and down it, freeze the transform, helpers to link to, and the deformers a rig starts with.
Archviz - Build	Architecture objects in 3ds Max's own creation modes: walls, doors, windows, stairs, railings and plants - drag them in the viewport as from the Create panel.

The same files are in the **Dials** folder of the download, for **Load dial...** as well.

The new-dial wizard

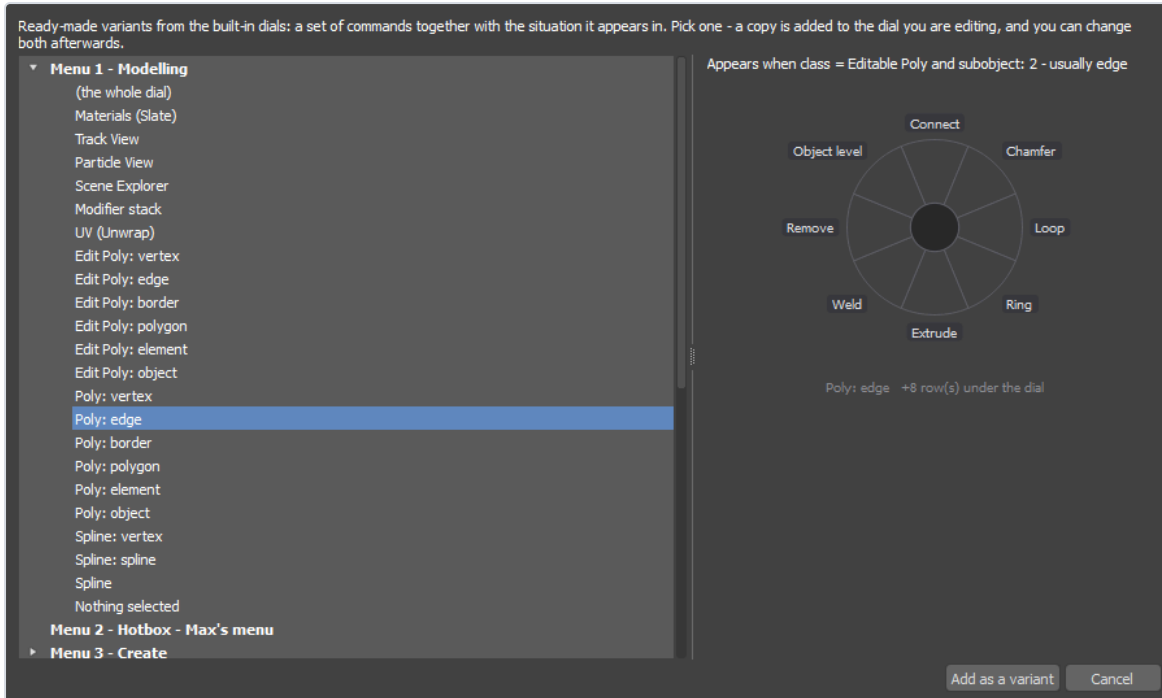
- 1 **New dial wizard...** under the slot list.
- 2 **Page 1** - which dial, its name, and whether the new contents become a **new set** (the default: what the dial shows now is kept) or **replace** it.
- 3 **Page 2** - eight fields, one per direction. Type part of a name and pick from the list: every 3ds Max command and the ready-made scripts. The drawing beside the fields follows; a name that matches nothing turns red and is left out.
- 4 **Page 3** - what will be made. Finish puts it in the editor; Save makes it real.



The wizard's second page.

Ready-made variants - From a template...

Every context variant of the built-in dials is a tested set of commands with its condition: Editable Poly and Edit Poly level by level, splines, cameras, lights, the Slate, Track View and Scene Explorer windows. **From a template...** lists them all, with each built-in dial as a whole; pick one and press **Add as a variant**. It arrives with its condition (a whole dial asks you for one) at the end of the variant list - move it up if a more general variant above would catch it first.



The templates: what they are on the left, when they appear and what they hold on the right.

14. A good dial - a checklist

- **Eight is the limit for the hand.** What you rarely use goes to the list or a submenu; a different task gets a set of its own.
- **Opposites face each other** - hide / unhide, grow / shrink, in / out.
- **The same command in the same place** on every variant: Extrude north on every level teaches itself.
- **Captions of one or two words.** The dial is read at a glance.
- **Keep the caddy past the rim** for commands whose settings you change only sometimes.
- **Watch the Usage column** after a week: a direction never used is a free place; one often cancelled is a command in the wrong place.
- **Save a preset** before big changes - Load preset brings everything back.

Still stuck? Open an issue at github.com/looki666/MarkingForge/issues or write to forgeplugins@gmail.com.

MarkingForge > Report a bug... puts your 3ds Max and MarkingForge details on the clipboard, ready to paste.

Support, bug reports and feature requests

ISSUES	github.com/looki666/MarkingForge/issues
E-MAIL	forgeplugins@gmail.com
WEB	https://forgeplugins.tech (forgeplugins.tech - more plugins by the author)

Bug reports and feature requests are best opened as issues on GitHub - the forms there ask for exactly what is needed, and you can follow the fix. Questions, and anything you would rather not post publicly (licence, purchase), go to the e-mail address. Every message is read.

Reporting a bug

- 1 In 3ds Max choose **MarkingForge > Report a bug...** - it copies your 3ds Max version, MarkingForge version, configuration summary and start-up log to the clipboard. It works even when the plugin did not load.
- 2 Open github.com/looki666/MarkingForge/issues and pick **Bug report** - or answer **Yes** in the dialog to write an e-mail to forgeplugins@gmail.com instead.
- 3 Say what you did, what you expected and what happened instead, then paste the details with **Ctrl+V**.
- 4 Add a screenshot if you can. When the problem is in one dial, attach a preset of it (editor > **Save preset...**).

Suggesting a feature

Open an issue with the **Feature request** form, choose **MarkingForge > Suggest a feature...**, or simply write. Describe what you would like to do and how you do it today - a concrete example from your work says more than a list of wishes. Ideas for new built-in dials, commands and options are all welcome.

Nothing is sent by the plugin itself: the details go to the clipboard and your own e-mail program, so you see everything that leaves your computer. MarkingForge has no network access.